CERFACS

CENTRE EUROPÉEN DE RECHERCHE ET DE FORMATION AVANCÉE EN CALCUL SCIENTIFIQUE

Groupe Advanced Aerodynamics and Multiphysics

Université de Toulouse

Habilitation à Diriger les Recherches

-

Sur des méthodes de CFD appliquées aux écoulements industriels

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www.cerfacs.fr

Outline

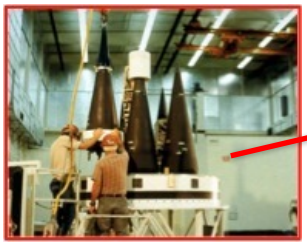
- Career description
- Research management
- Industrial ``state of the art'' and future applications
- Description of former research activities
- Current and future research areas

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Work experience (1)

- 1998-2001: PhD Thesis at Université de Montpellier II (Adv. B. Mohammadi)
  - Wall function for RANS simulations of hypersonic flows over smooth and rough adiabatic and isothermal walls
  - Contract between CEA/CESTA and U. de Montpellier II
- 2001-2003: Post-doctoral position at CERFACS on wake vortices
- 2003-2005: research engineer on military/civil reentry flows.
  - Hypersonic and high enthalpy flows including multi species chemistry at non equilibrium and radiation. Wall functions. Code development, validation and best practices.





Test from SNLE Le Terrible
(issued from blog RPDefense - 06/04/2013)

Work experience (2)

- Since 2005: Senior researcher
  - Research and software development for CERFACS shareholders
    - Involvement in the *elsA* code of ONERA
    - Prototype of a cell-centred unstructured Finite Volume solver for implementing and testing schemes
    - Since 2012, large involvement in the development of *JAGUAR*, a high order CFD code based on the Spectral Difference paradigm









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Scientific activities

- CERFACS activities:
  - Organizer of the scientific seminar entitled ``***Wall and LES***'' extended to any ingredient for LES - 8 meetings/year
  - Lecturer and organizer of one CERFACS training sessions dedicated to LES
- Reviewer:
  - *Journal of Computational Physics*
  - *AIAA Journal*
  - *ASME Turbo Expo* conference
- Administrative tutor for MEGEP PhD school
  - Evaluation of any human / technical problem

Research projects

- Involvement in many European projects:
  - SimSAC Simulating Aircraft Stability and Control Characteristics for Use in Conceptual Design: wall functions and ALE
  - AeroTraNet2 AERONautical TRAining NETwork in Aerodynamic Noise from Widebody Civil Aircraft): supersonic jet noise
  - JERONIMO Jet noise of High Bypass Ratio Engine: Installation, Advanced Modelling and Mitigation: acoustic sources with jet / wing interactions
  - AFLoNext 2nd Generation Active Wing Active Flow- Loads & Noise control on next generation wing): unsteady pressure force on MLG doors
  - DPAC and Aerospace Valley Projects, DGCIS funding
  - Direct contracts with our shareholders

Research network

- Main academic collaborations on HPC, numerics and physics:



















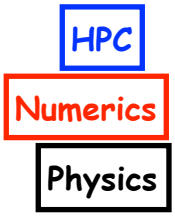












- Industrial collaborations with:







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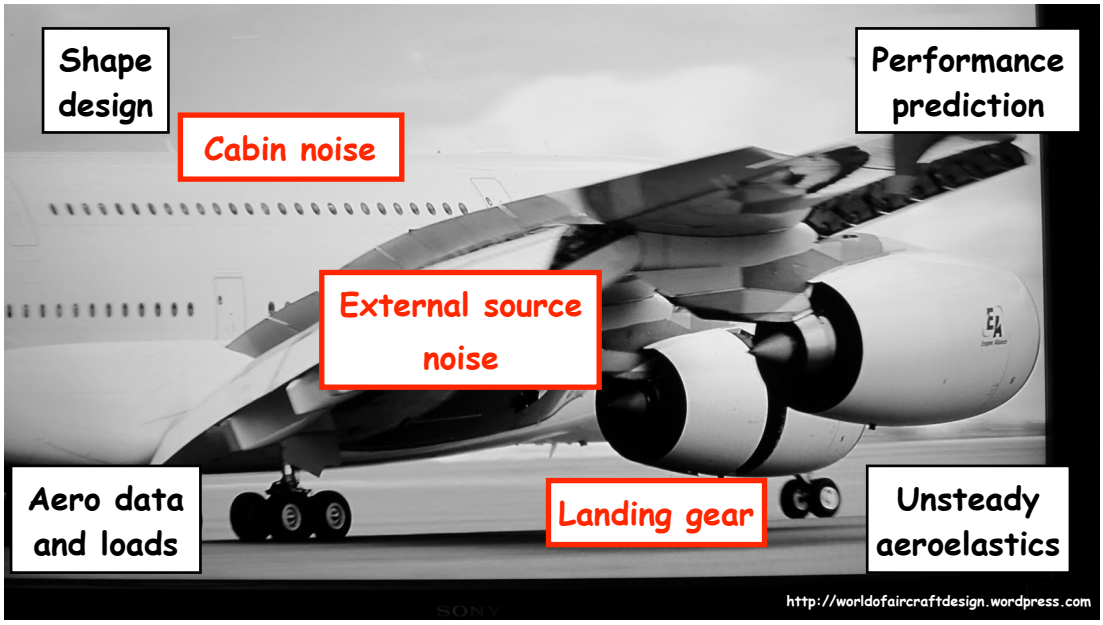
CFD simulations at Airbus

- 4 main areas:



CFD simulations at Airbus

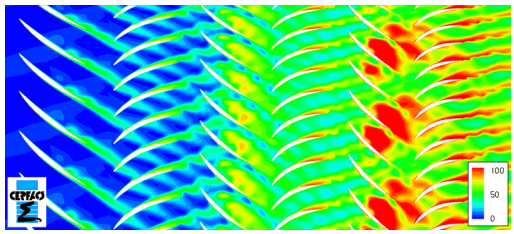
- 4 main areas:



CFD simulations at Safran

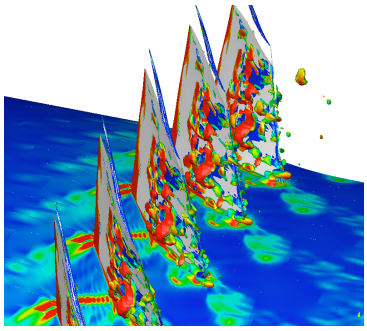
- Multistage turbomachine computed with mixing plane approach

Wlassow, Gourdain and Montagnac
URANS comp. near stall, 134 Mcells
40 days on 4096 cores (IBM BG /P - EDF)



- Tendency:
  - Current: Introduction of unsteady effects with URANS computations on 3D full configuration
  - Future: Replace (U)RANS by LES type approach
  - Account for complex phenomena (transition…)
  - Go further in taking into account the true (complex) geometry
  - Multi-physics simulations (aerothermal, aeroelastic…)

Gomar, Gourdain and Dufour
LES on rotor 37, 100 M cells
30 days on 512 cores (IBM BG /L)



Conclusion on industrial needs

STATUS:

- RANS/URANS, geometry not too complex, structured grid

FUTURE:

- For **turbomachinery: URANS simulations** to capture multi-stage unsteady effects
- For **aircraft and turbomachinery:** treat off-design configurations with **LES**

- **Complex geometry** - unstructured / Cartesian grid
- **Turbulent fluctuations** to capture and transport
- **Multi-physics**

Conclusion on industrial needs

STATUS:

- RANS/URANS, geometry not too complex, structured

FUTURE:

=> Focus on URANS simulations for recent research!

=> Focus on LES for future work!

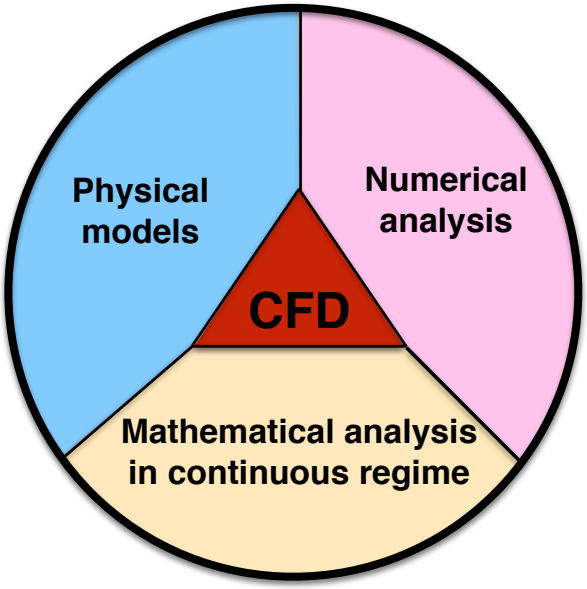
- ...ure and transport
- ...ysics

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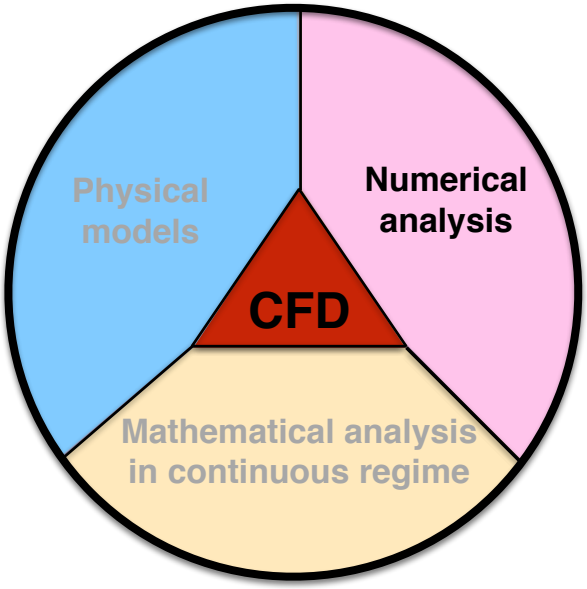
Research thematics

- **RANS turbulence models and wall functions**
- **Mathematical aspects of RANS turbulence models**
- **Discrete analysis of the Navier-Stokes equations**



Research thematics

- **RANS turbulence models and wall functions**
- **Mathematical aspects of RANS turbulence models**
- **Discrete analysis of the Navier-Stokes equations**
  - **HBT: technique to decrease CPU time of periodic and almost periodic flow simulations**



Harmonic Balance Technique (HBT)

- Many industrial flows are periodic in time (T):
  - Turbomachinery flows, analysis of unsteady force response to pitching airfoil with the frequency issued from structural analysis…
- By nature, Euler equations are hyperbolic
  => Transient regime must be by-passed, during 0.5T to many T
- Harmonic Balance Technique developed by Duke and Stanford University (Gopinath *et al*, 2007)
  - **New time integration procedure for *PERIODIC* URANS computations!**
  - **Way to decrease CPU time for computations!**
  - **Period must be *KNOWN*!**
- At CERFACS, approach analyzed and implemented during PhD thesis of F. Sicot, T. Guédeney and A. Gomar

Principle (1)

- Start from the discretized equations (FV): $V\frac{dW}{dt} + R(W) = 0$
- Assume that both **flow and residual are periodic**:

$$W = \sum_{k=-\infty}^{+\infty} \widehat{W}_k e^{ik\omega t} \qquad R(W) = \sum_{k=-\infty}^{+\infty} \widehat{R(W)}_k e^{ik\omega t}$$

- Introduce both relations and use the **orthogonal Fourier basis**:

$$ik\omega V\widehat{W}_k + \widehat{R(W)}_k = 0 \quad \forall k$$

- CFD solvers always solve equations in the ***time domain***: for simplicity, it is needed to transfer the equations in the time domain
  - Use the ***Inverse Fourier Transformation Matrix - IFTM***

Principle (2)

- Two ways to consider the IFTM:
  - With an ***odd*** number of Fourier coefficients
  - With an ***even*** number of Fourier coefficients

- Mandatory to choose **an odd number of Fourier coefficients for stability: 2N+1 coefficients Fourier coefficients!** Gopinath, Jameson, AIAA Paper 2006

- From now on, select a limited number of coefficients $-N \leq k \leq N$ and introduce the IFTM:

$$ik\omega V\widehat{W}_k + \widehat{R(W)}_k = 0 \quad -N \leq k \leq N$$

$$\overline{R}(\overline{W}) + S(\overline{W}) = 0, \quad \begin{cases} \overline{W} = (W_{t=0}, ..., W_{t=2N\,T/(2N+1)}) \\ \overline{R}(\overline{W}) = (R(W)_{t=0}, ..., R(W)_{t=2N\,T/(2N+1)}) \end{cases}$$

Principle (3)

- Starting from:

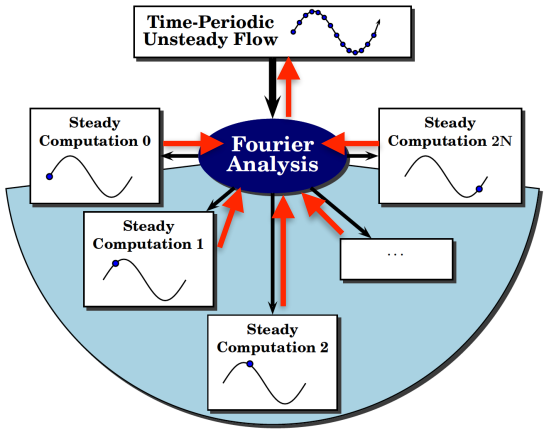
$$\overline{R}(\overline{W}) + S(\overline{W}) = 0, \quad \begin{cases} \overline{W} = (W_{t=0}, ..., W_{t=2N\,T/(2N+1)}) \\ \overline{R}(\overline{W}) = (R(W)_{t=0}, ..., R(W)_{t=2N\,T/(2N+1)}) \end{cases}$$

- We introduce a time marching to solve equations with standard solvers:

$$V\frac{d\overline{W}}{d\tau} + \overline{R}(\overline{W}) + S(\overline{W}) = 0$$

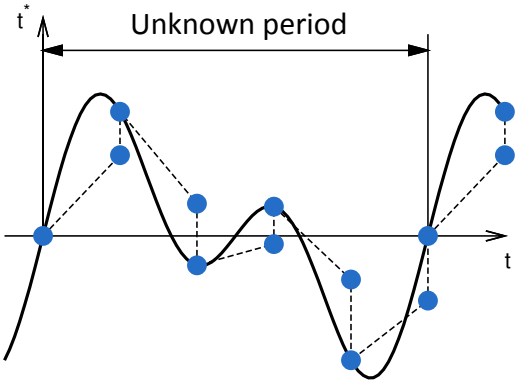
- Explicit formulation of source term:

$$S_n = \frac{2\pi}{T}\sum_{m=-N}^{m=N} d_m W_{n+m}$$

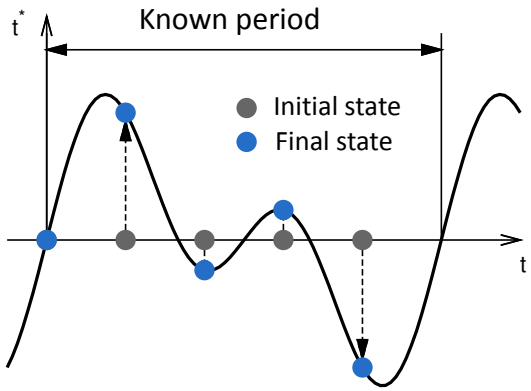


Principle (4)

- Summary:



- Present influences future
- Parallel in space
- Unsteady transient regime
- Memory cost ≈ RANS comp.
- Computation cost > 100 RANS comp.



- All time instants are coupled
- Parallel in space AND time
- Stationary convergence
- Memory cost ≈ (2N+1) RANS comp.
- Computation cost ≈ (2N+1) RANS comp.

Impact of HBT on time step

- Classic stability analysis on Euler equations in 1D:

$$\delta\tau = CFL\frac{L}{\|\vec{u}\| + c}$$

- Gopinath and Jameson perform a stability analysis for HBT:

$$\delta\tau = CFL\frac{L}{\|\vec{u}\| + c + \omega NL}$$

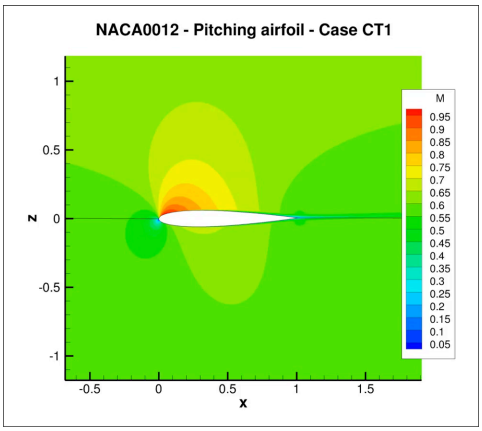
  - Largest stable time step is reduced for a large number of harmonics
  - Largest stable time step is reduced for a large frequency

- **Explicit time-integration approach is too CPU time-consuming!**

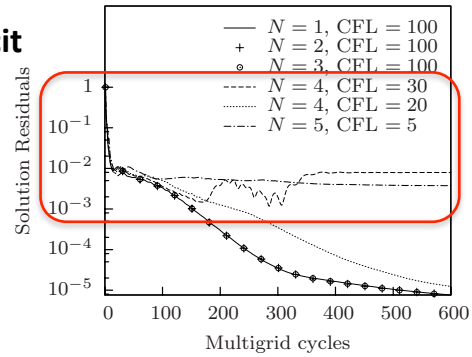
Implicit formulation

- Simplest approach: linearize the source term
  - Matrix links all time steps for any mesh cell
  - Needs to rewrite implicit formulation of any CFD code since implicit treatment is defined on ONE block and NOT over MANY blocks
- Development of new implicit treatments:
  - No need for the full implicit matrix of source term
  - Method BJ-SSOR (Block-Jacobi Symmetric Successive OverRelaxation): fast but weakly robust (2 updates for 2 sweeps)
  - Method BJ-SOR (Block-Jacobi Successive OverRelaxation): robust but larger CPU cost (update at any sweep)
- ***First publication on implicit treatment***
  - F. Sicot, G. Puigt and M. Montagnac, Block-Jacobi Implicit Algorithm for the Time Spectral Method, *AIAA J.*, 2008, 46(12), 3080-3089
    - Cited more than 14 times

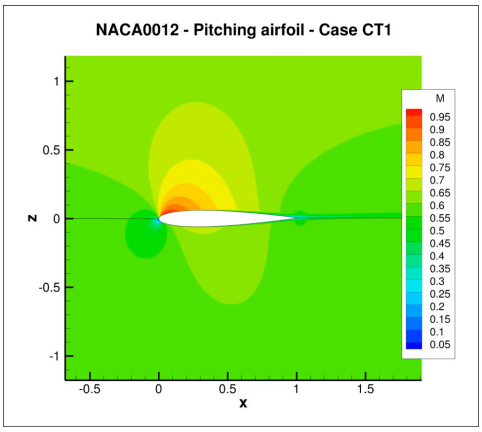
Validation - External aerodynamics - academic



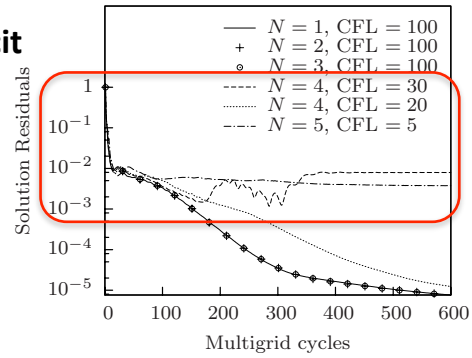
Before implicit treatment!



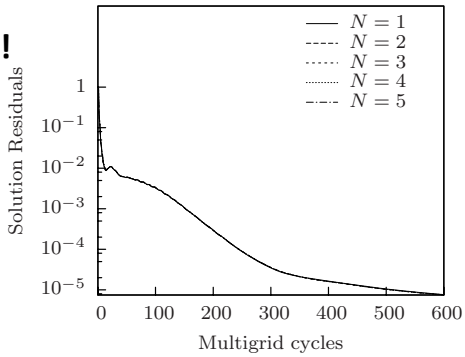
Validation - External aerodynamics - academic



Before implicit treatment!

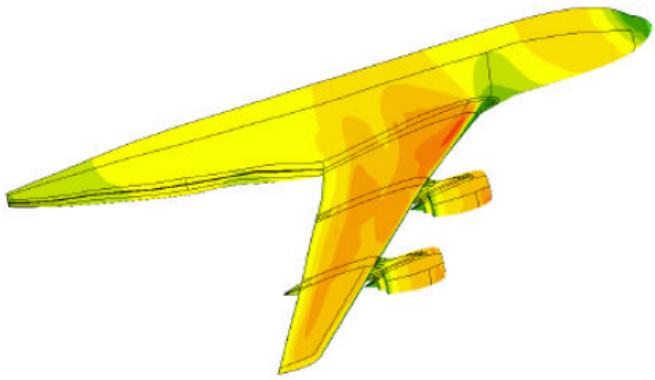


Convergence improvement!



Validation - External aerodynamics - industry

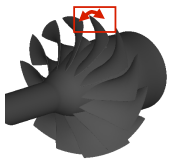
- A380 configuration: transonic flow

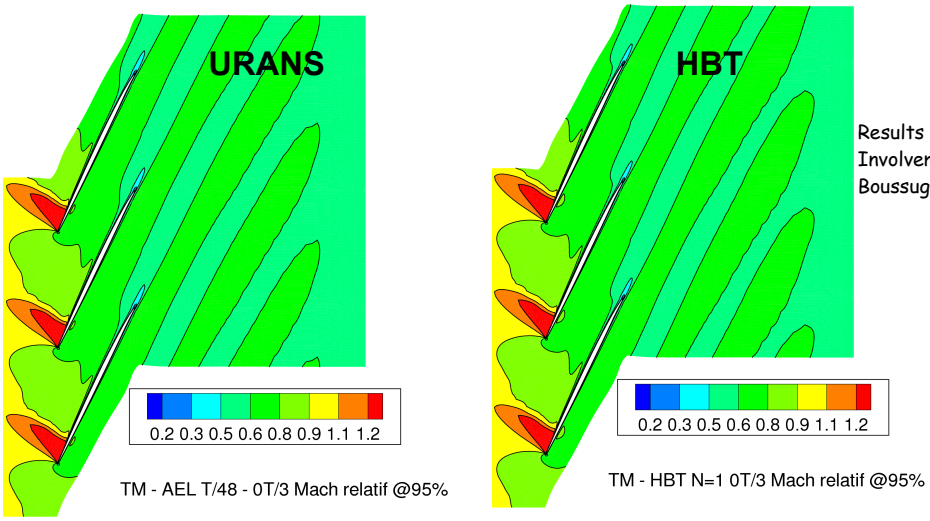


- **CPU time gain between 2 and 6 for using HBT instead of URANS!**
- **Discrepancy between URANS and HBT can be found…**
  … due to a bad management of URANS computation!!!

Validation - Confined flow

- Example of validation results on aeroelasticity for turbomachinery: CPU time gain between 4 and 10!
- Discrepancy between URANS and HBT can be found…
  … due to a bad management of URANS computation!!!

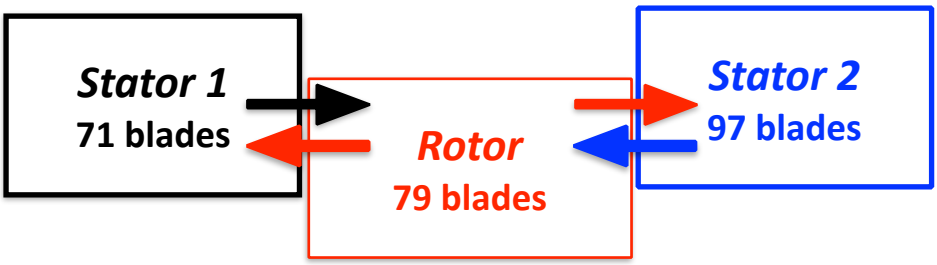




Results from MACAO Project, Involvement of Dufour, Sicot, Boussuge, Puigt

Need for a multi-frequential approach

- For turbomachinery, it is relevant to change the referential frame to the one associated with the blade:
  - ``Static'' blade seeing the effects of surrounding ``mobile'' blades



- **In Stator 1 and Stator 2, unsteady effects due to Rotor movement at BPF_R (Blade Passing Frequency of Rotor)**
- **In Rotor, unsteady effects due to Stator 1 and Stator 2 movements at BPF_{S1} and BPF_{S2}**
- **Two frequencies spectral content if $BPF_{S2} / BPF_{S1} \neq k,\ k \in \mathbb{N}$**

New set of equations

- The process to define the equations is almost the same as for one-frequency HBT.
- Start from the discretized equations (FV): $V\frac{dW}{dt} + R(W) = 0$
- Select a list of frequencies: $(\omega_0 = 0, \omega_1, \ldots, \omega_N)$
- Projection of variables and residuals on the set of frequencies:

$$\sum_{k=-N}^{k=N} (i\omega_k V\widehat{W}_k + \widehat{R}_k)e^{i\omega_k t} = 0$$

- Choose *2N+1* time instants such that $\mathcal{A}_{k,n} = exp(i\omega_k t_n)$ is ***invertible***
- Apply the inverse Fourier Transform to the system: $V\frac{d\mathcal{A}}{dt}\mathcal{A}^{-1}W^* + R^* = 0$ with:

$$W^* = (W(t_0 = 0), \ldots, W(t_{2N}))$$
$$R^* = (R(t_0 = 0), \ldots, R(t_{2N}))$$

and add a pseudo-time term to time march equations.

Differences between single and multiple frequencies

One frequency

- 1 frequency and its harmonics
- Transformation matrix is invertible by nature
- Uniform sampling of time instants over the period
- Direct expression of the matrix coefficients (real by nature)

$$d_{j,l} = \frac{(-1)^{j-1}}{2\sin\left(\frac{\pi(j-1)}{2N+1}\right)}$$

Several frequencies

- Predefined frequencies
- Transformation matrix NOT necessary invertible
- Time instants chosen to make the transformation matrix invertible
- Coefficients to compute numerically (imaginary part very small 10^{-14})

$$\frac{d\mathcal{A}}{dt}\mathcal{A}^{-1}$$

First results

- Problem:
  - Strong influence of the time instants on the ***stability*** and ***convergence***!
  - ***Simple choice*** of time instants ***doesn't work!***
  - Never seen before in literature!

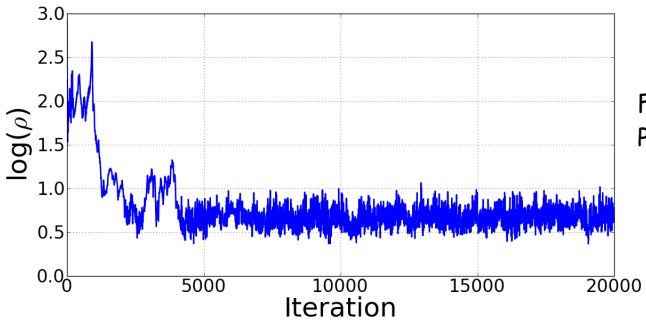


Figure from Guédeney PhD defense

- Toy problem shows that the condition number of the matrix in the source term drives accuracy and stability of the multifrequency HBT
  - Frequencies fixed, time instants=parameters!

Correction techniques

- ***APFT approach*** (dev. by Kundert *et al.* in electronics):
  - Maximize orthogonality of vectors by Gram-Schmidt orthogonalization
  - Strong improvement in condition number but not enough for industrial use
- ***OPT algorithm*** (**New!**):
  - *Strategy:* minimizing condition number = optimum problem
  - Use a L-BFGS-B approach to obtain a (local) minimum
  - Key point is the definition of the initial state for L-BFGS-B
    - Very refined sampling = initial guess
    - Compute condition number for all guess and apply then optimization algorithm
    - Find the minimum condition number
  - **Strategy works fine since the OPT algorithm has a very low CPU cost**

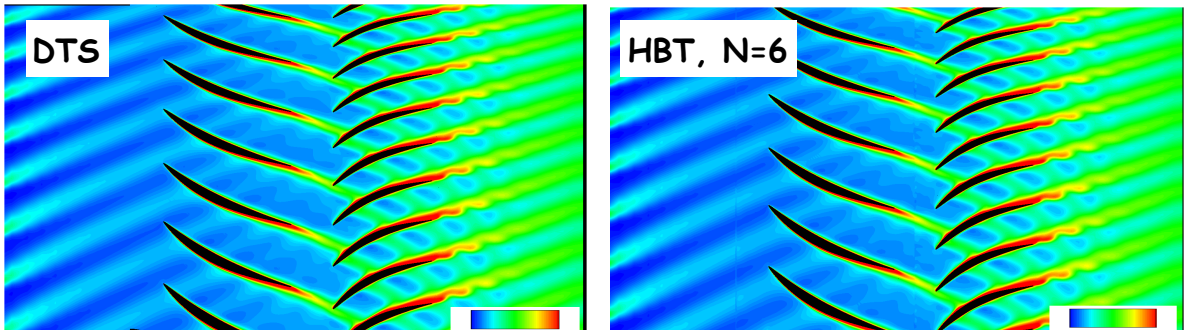
Results

- Example of improvement:

Condition number	Min	Max	Mean
Uniform sampling on the largest period	*1*	*9.4 x 10*	*1.5 x 10*
OPT	*1*	*2.6*	*1.1*

- Work applied on turbomachinery flows: first rows of CREATE Entropy fields DTS / HBT - Entropy

From Guédeney, Gomar, Gallard, Sicot, Dufour and Puigt, JCP, 2013



Conclusions on HBT

- New technique for periodic and almost periodic simulations
- Convergence is better monitored than with URANS approach
- Two key points addressed:
  - Implicit treatment for an enhanced convergence
  - Multi frequential stability and conditioning
- Results are encouraging and approach is available in *elsA*

- Still one question to address:
  - Harmonic content is full in the presence of a strong discontinuity
  - Harmonic transform approach needs many harmonics for accuracy and URANS can become more efficient
  - Analysis of HBT capability *a priori*, following Gomar, Bouvy, Sicot, Dufour, Cinnella, and Francois, JCP, 2014.
  - Way to overcome the limitation?

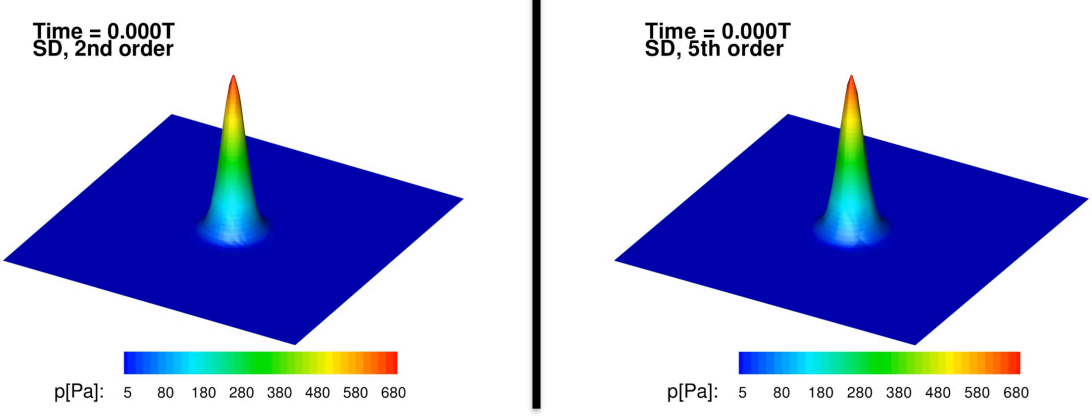
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- Short overview of ``the state of the art''
- Description of former research activities
- Research management
- Current and future research areas

Current and future research (1)

- Current and future research concerns **LES** at **High Re** on **non academic configurations**
- First point to address: the choice of schemes
- Two alternatives: low order or high order schemes

Low order ***High order***



Current and future research (1)

- Current and future research concerns **LES** at **High Re** on **non academic configurations**
- First point to address: the choice of schemes
- Two alternatives: low ord

Better spectral behavior with HO schemes:
- Less dissipation and less dispersion
- Same accuracy attained with less degrees of freedom with HO scheme (64 time more mesh nodes to attain the same accuracy with a second order scheme!)



Current and future research (2)

- Non academic configuration: ***very complex geometry***
  ➥ *Mandatory to consider **unstructured grid!***

- ***Many grid points*** and ***High order schemes*** - **O(10^9) DoF**
- ***Explicit time integration*** with ***small time steps*** (LES status)
  ➥ ***High Performance Computing capability!*** - **O(10^6) cores**

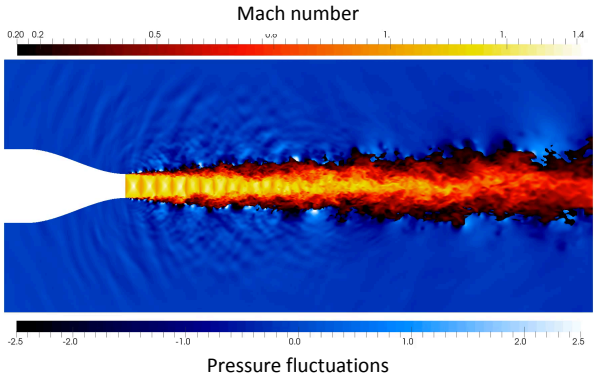
- **Two ways to address the problem:**

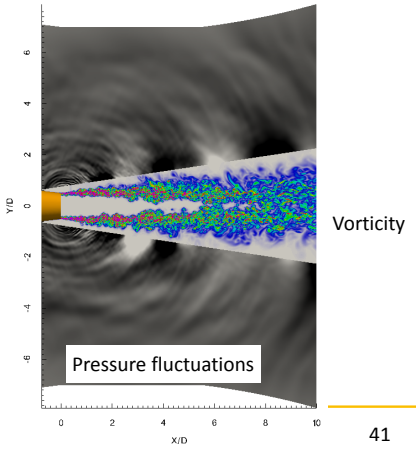
➥ ***Using standard discretization technique (Finite volume)***

➥

Approach 1: context

- For structured grids:
  - Industrial demand to compute acoustic sources with a LES approach
  - Definition, implementation, validation of a high order compact scheme adapted to Finite Volume developed during PhD of A. Fosso-Pouangué + dedicated boundary conditions
  - Application to aeroacoustic jet simulation in subsonic and supersonic regimes

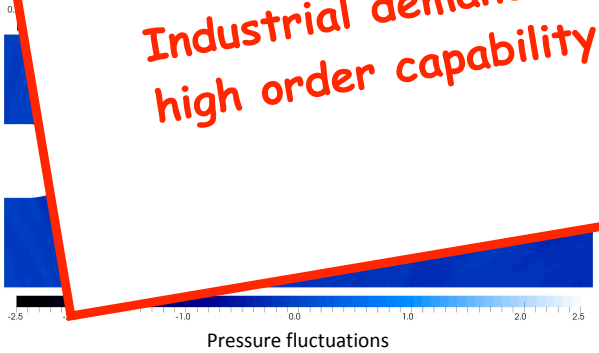


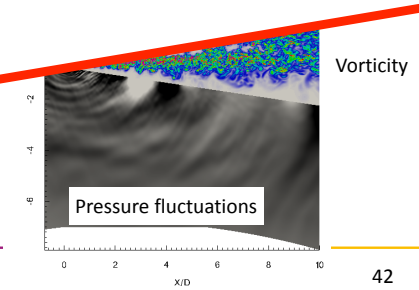


Approach 1: context

- For structured grids:
  - Industrial demand to compute acoustic sources with a LES approach
  - Definition, implementation, validation of a high order

Industrial demand to develop and implement high order capability for unstructured grids

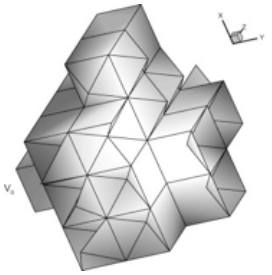




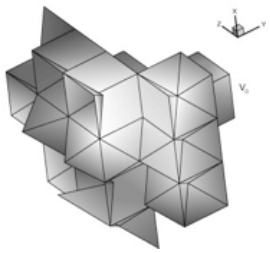
Approach 1: LES and FV schemes (1)

- Standard High Order schemes follow a polynomial approximation to define an interface value
  - Define the necessary stencil
  - Apply an ENO/WENO/K-exact formulation on the stencil

- Large stencil is bad for HPC:



5th order - 104 cells



6th order - 167 cells

Figure from Tsoutsanis, Titarev and Drikakis, JCP, 2011

Approach 1: LES and FV schemes (2)

- **For LES of non academic configurations, the key point is the definition of a convection scheme with an order > 2**

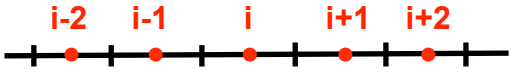
- Choice to focus on an approach with a limited stencil:
  - Directional approach that involves the use of states and state gradients at cell center and mesh interface
  - Less DoF
  - Not exact in the sense that the true HO scheme order cannot be guaranteed on all element shape

- Work performed during PhD thesis of P. Cayot (CIFRE Snecma)

Approach 1: description (1)

- 1D analysis of Euler equation: $\frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} = 0$
- FV approach to define numerical flux $\mathcal{F}$ in cell *i*

$$\mathcal{F} = \frac{1}{\Delta x}(\Phi_{i+1/2} - \Phi_{i-1/2})$$



- Interface flux $\Phi_{i+\frac{1}{2}} = \frac{f^-_{i+\frac{1}{2}} + f^+_{i+\frac{1}{2}}}{2} + \delta \frac{f^-_{i+\frac{1}{2}} - f^+_{i+\frac{1}{2}}}{2}$ with dissipation parameter
- Directional upwinding: $\begin{cases} f^-_{i+\frac{1}{2}} = f_i + \frac{1}{2}\vec{\nabla} F^-_{i+\frac{1}{2}} \\ f^+_{i+\frac{1}{2}} = f_{i+1} - \frac{1}{2}\vec{\nabla} F^+_{i+\frac{1}{2}} \end{cases}$

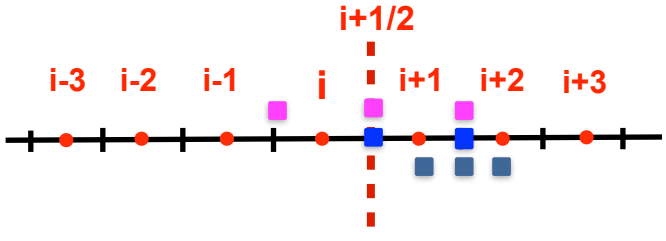
Approach 1: description (2)

- Two gradients: Interface and Cell-Centered Gradient in 1D:

$$\vec{\nabla} F_{i+1/2} = \frac{f_{i+1} - f_i}{\Delta x} \qquad \vec{\nabla} F_i = \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$

- Build a slope with these gradients:

$$\vec{\nabla} F^+_{i+\frac{1}{2}} = (1-\beta)\vec{\nabla} F_{i+1/2} + \beta \vec{\nabla} F_{i+3/2} + \theta(\vec{\nabla} F_{i-1/2} - 2\vec{\nabla} F_{i+1/2} + \vec{\nabla} F_{i+3/2}) + \xi(\vec{\nabla} F_{i+1} - 2\vec{\nabla} F_{i+3/2} + \vec{\nabla} F_{i+2})$$



Conclusion on approach 1 (1)

- Context: Finite Volume discretization
- *Status - Structured scheme:*
  - Compact scheme dedicated to aeroacoustic including shocks
- ***Research - Unstructured scheme:***
  - **Directional** approach, **compact** stencil, status: finalization and first test cases
- Then, towards ***mesh flexibility*** and ***efficiency***:
  - **Coupling of structured and unstructured approaches** inside ONE CFD kernel (elsA)
  - Coupling strategy based on **Nonconforming Grid Interface**
    - Computation of flux on facets
  - Spectral analysis for NGI + high order schemes

Conclusion on approach 1 (2)

- In the near future, research will focus on:
  - LES on complex geometry and at high Reynolds number
  - Massively parallel computations with **O(10^6) cores** and **O(10^9) degrees of freedom**
  - Current practice: Explicit time integration
- Stencil is compact BUT not fully local
- Interest in analyzing a new approach with **a compact stencil** and **very good properties for HPC (parallel issues)**

Current and future research (2)

- Non academic configuration: ***very complex geometry***
  ➥ *Mandatory to consider* ***unstructured grid!***

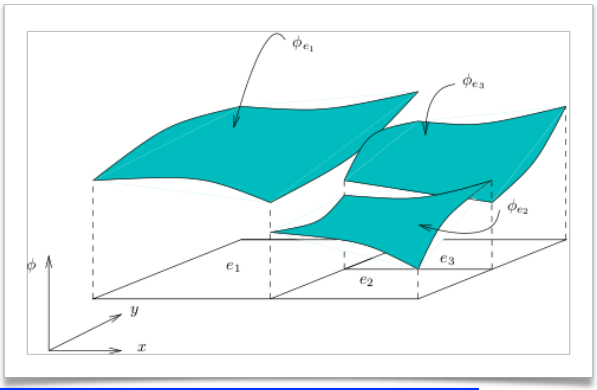
- ***Many grid points*** and ***High order schemes*** - **O(10^9) DoF**
- ***Explicit time integration*** with ***small time steps*** (LES status)
  ➥ ***High Performance Computing capability!*** - **O(10^6) cores**

- **Two ways to address the problem:**
  ➥ *Using standard discretization technique (Finite volume)*
  ➥ ***Using recent discretization technique (Spectral Difference)***

Approach 2: Spectral difference approach (1)

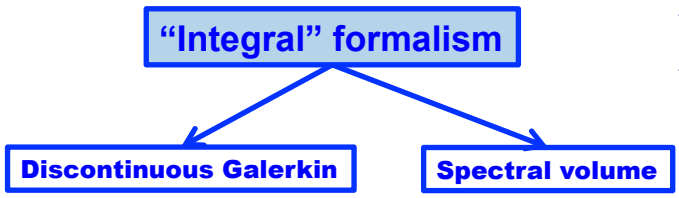
- Even using a technique for a compact stencil, it is not sure that classic FV high order schemes will lead to HPC capabilities
- **Analyze and test a new paradigm: projection of equations on the set of polynomials of degree *p***

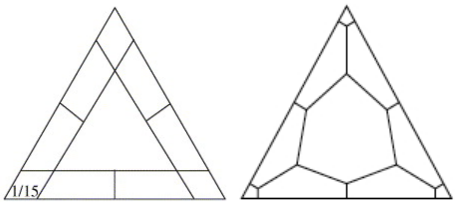
Choice to focus on High Order Discontinuous (HOD) methods for unstructured grids



Locally continuous BUT Globally discontinuous!
Compact stencil!

Approach 2: Spectral difference approach (2)



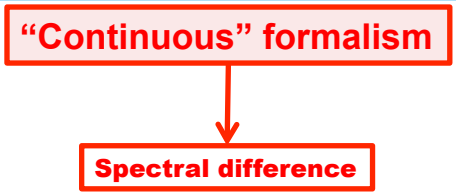


- DG based on FE per cell
- SV based on FV on sub cells
- Both approaches need quadrature rules to compute integrals

- **DG** leads to **strong constraints on time step**
- **SV** depends on **subcell definition**: clear on tetrahedra only

- **New techniques have shown to be more efficient**

Approach 2: Spectral difference approach (3)



- SD based on solution of strong equations:
  - Avoid complex quadrature rules!
- Only available on hexahedra
- First paper: Kopriva 1996 for structured grids
- Then: Liu, Wang and others in 2006 for unstructured grid

- Many work performed during the last 10 years:
  - Many European projects (ADIGMA, IDIHOM): focus on same quality results as classic FV solvers using less DoF - RANS approach
  - But the use of these schemes for LES is quite new and the number of papers is (very!) limited!

Approach 2: Spectral difference approach (4)

- ***Key points on SD:***

$$\frac{\partial W}{\partial t}\Big|_i + \frac{\partial \mathcal{F}(W)}{\partial x}\Big|_i = 0$$

  - **Strong form** of equations to solve
  - Staggered approach with **solution points** / **flux points**

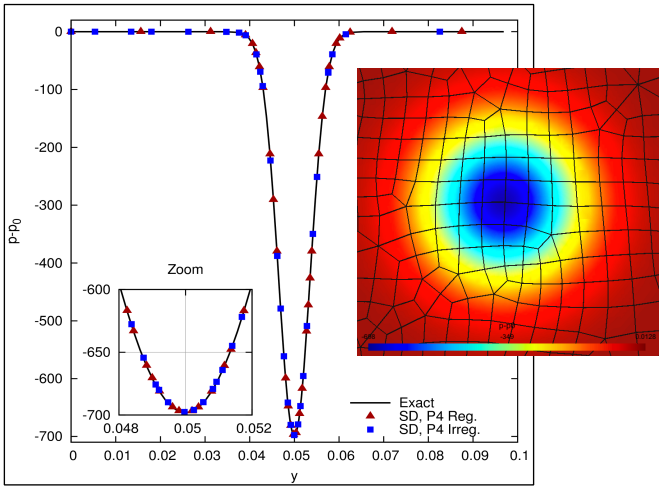


  - **Conservative** by nature (same polynomial for two elements sharing a face)
  - **Simple** to understand but efficient!
  - Key point: continuous flux with a **Riemann solver** at interface
  - **Diffusion scheme** based on a **centered** approach

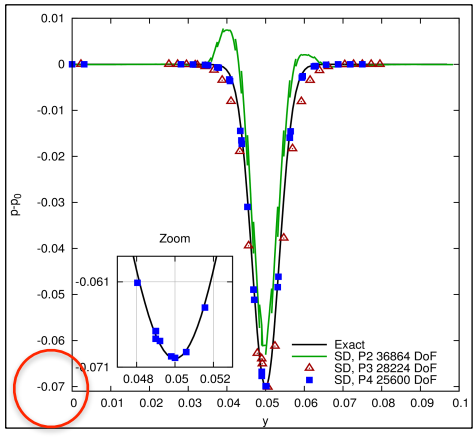
JAGUAR solver

- 2 year-old code in fortran 90
- High parallel performance

Isentropic COVO - 50 T_c
p_0 =101 000Pa, M=0.5



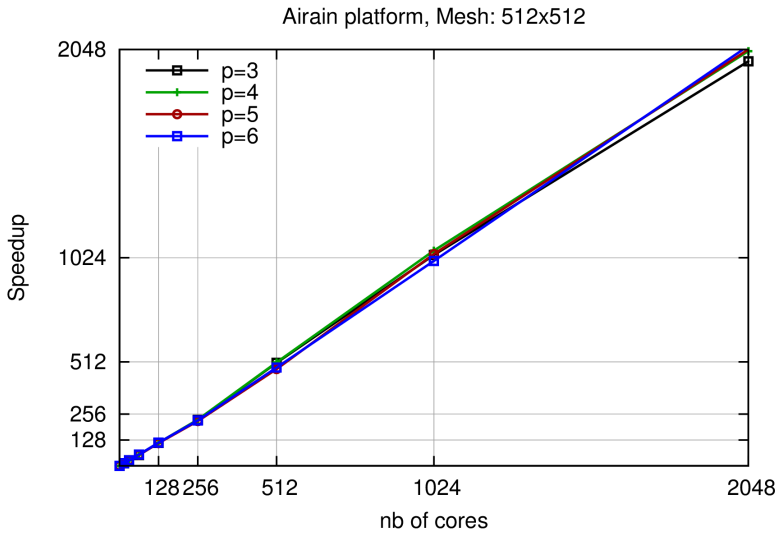
- Euler, NS and (I)LES
- High accuracy even when mesh lines are not aligned with flow physics



Isentropic COVO - 50 T_c
p_0 =101 000Pa, M=0.05

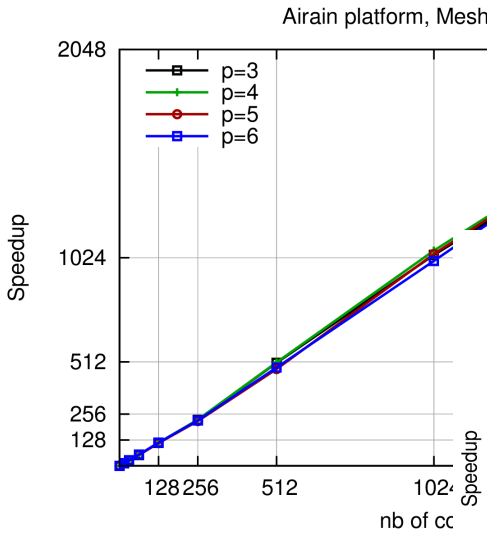
- Work with N. Villedieu (Post-Doc), J-F. Boussuge, A. Cassagne (Engineer CINES) and training students

JAGUAR efficiency



Almost 100% of efficiency up to 2048 cores

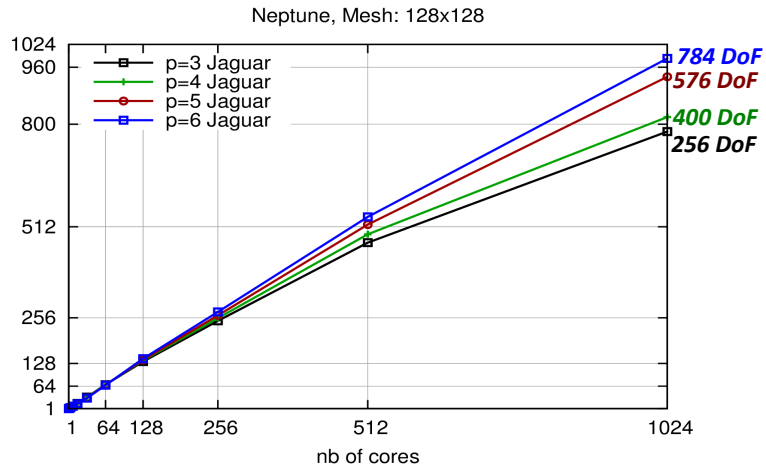
JAGUAR efficiency



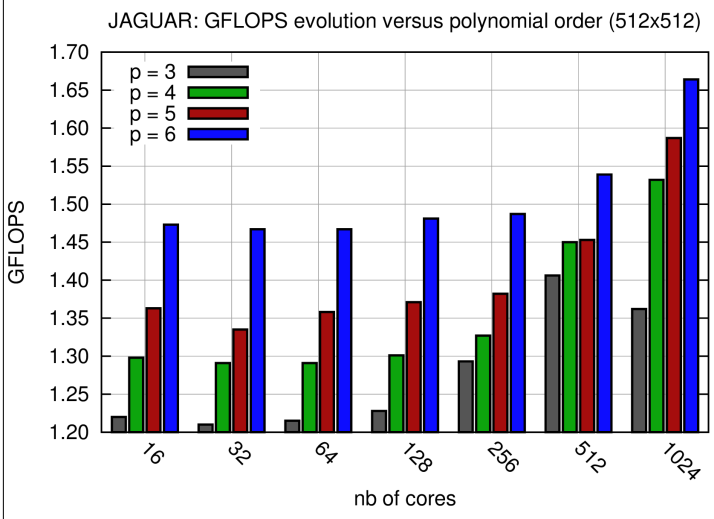
Almost 100% of efficiency up to 2048 cores

Still 80% of efficiency with only 400 DoF per core

p=4 ⇔ 125 DoF/cell

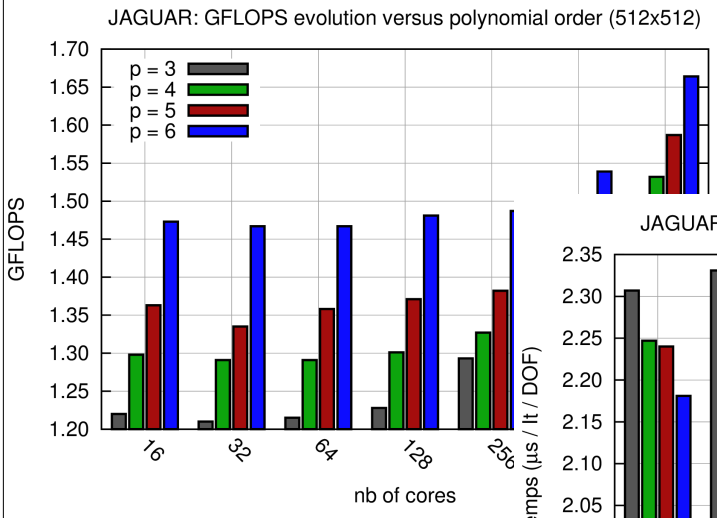


JAGUAR CPU performance



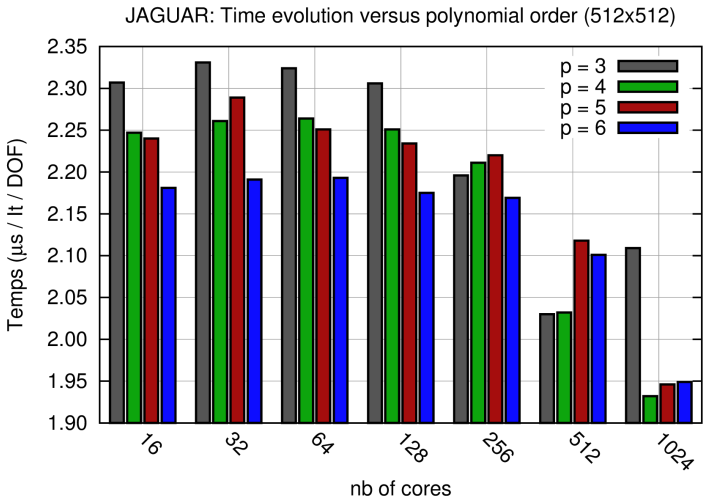
GFLOPS obtained are almost twice as big as standard CFD solvers

JAGUAR CPU performance



GFLOPS obtained are almost twice as big as standard CFD solvers

Mean time by iteration and by DoF : 2.2 microsecond which is lower than standard CFD solvers



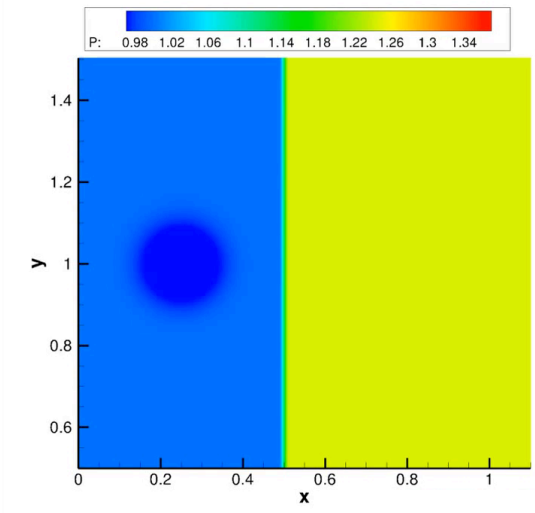
Status in high order method scientific community

Advanced approaches applied to LES:

- ***HiFiLES***: Open source code developed by **Stanford University** - SD-like approach - Cross comparison on COVO gives comparable results but *JAGUAR* is 8 times faster
- ***Argo-DG***: DG code developed by **CenAero** - best European solution regarding HOM - PRACE projects… - *JAGUAR* is about 2 times faster
- ***AGHORA***: DG code developed by **ONERA** - Not fairly compared with *JAGUAR* at present time
- …

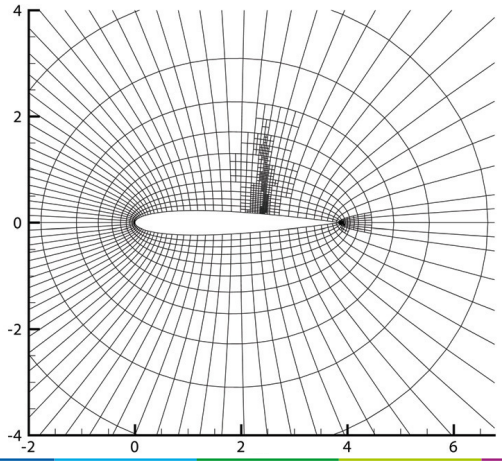
JAGUAR: work to publish

- ***Handling discontinuity*** with M. Lemesle and N. Villedieu:
  - Linear reconstruction
  - Add a diffusion term (constant or optimized - adapted from Persson and Peraire, AIAA P. 2006-0112)
  - **Piecewise Parabolic Method:** adapted from Colella and Woodmard
  - **Multidimensional Optimal Order Detection**: local decrement of polynomial degree where a solution is not ``acceptable'', adapted from Loubère, Diot, Dumbser and Clain



Research to address in the future (1)

- ***Extension of SD method:***
  - SD on **triangle** and on **tetrahedra, prisms, pyramids**
  - ***hp*-adaptation**, including **hanging-nodes** and **curved elements**
  - Analysis of transport, positivity, boundedness, stiff source terms for **multi species reacting flow**



Li, Premasuthan et Jameson, AIAA Paper 2010-4435

Research to address in the future (2)

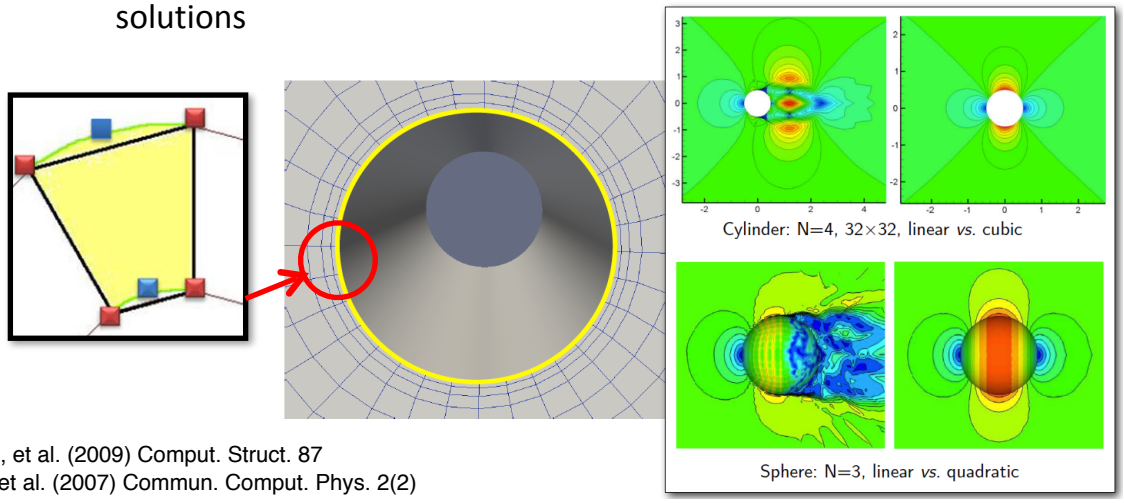
- ***LES dedicated treatments:***
  - **Wall treatment:**
    - Wall-resolved approach + implicit time integration: avoid strong time step limitation due to mesh refinement near the wall
    - Wall functions
  - **Time integration:**
    - Local time stepping with *hp* adaptation and SD
  - **Subgrid model:**
    - Non uniform DoF distribution, anisotropic ➥ New subgrid models? New paradigm? Mimic subgrid scale dissipation by a filter built with mathematic and physic considerations?

Research to address in the future (3)

- ***New models for complex physics:***
  - **Combustion**, including all combustion models
  - Analysis of the capability of such an approach to perform **aeroacoustic** simulations and compare this approach with *elsA*
  - **Aeroelasticity with large structure variations:** diminish the risk of reversed cell with large elements, especially for structures with a non linear behavior

Research to address in the future (4)

- ***Mesh generation:***
  - **Curvature for mesh elements**: Faceted coarse grids lead to false solutions



Liang, et al. (2009) Comput. Struct. 87
Sun, et al. (2007) Commun. Comput. Phys. 2(2)

High order mesh elements mandatory on curved boundary:
If not, the discretization sees discontinuity in shape!

Many thanks to my (current!) colleagues!

- ***Scheme team:*** Pierre C., Julien V., Thomas G., Frédéric S., Guillaume D., François G., Hugues D., Christophe C., Jean-Christophe J., Pierre S.
- ***JAGUAR Team:*** Nadège V., Adrien C., Isabelle M., Marie L., Jean-François B., Hugues D., David DOA
- ***Aeroacoustic team:*** Sophie LB., Carlos PA., Guillaume D., Romain B., Jean-Christophe G.
- ***The local elsA team:*** Marc M., Michel G., Hugues D.
- ***The scientific discussion team:*** Thibault B.-B., Gaelle M., Yannis S., Alessandro M., Danilo DS., Jens-Dominik M., Jérôme DL., Jérôme D., Gabriel S.,…
- ***And all other scientists, engineers, administrative + computer support members at CERFACS!***

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Thank you for your attention!

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